AgenticData: A Universal Data Comprehension Engine with Schema Inference, Lineage Tracking, and Quality Assurance

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Abstract

Data processing tools solve individual format problems—CSV parsers read columns, SQL tools query tables, PDF extractors pull text—but none provides unified *comprehension*: understanding what data means across formats, tracking where it came from, and assessing whether it can be trusted. We present AgenticData, a universal data comprehension engine that addresses three fundamental problems in data management: *schema opacity* (not knowing what fields represent), *lineage blindness* (not tracking data provenance), and *quality silence* (not detecting degradation). The system ingests data from 16 formats with automatic detection at 90–95% confidence, infers schemas with 19 semantic field types (detecting emails, dates, currencies, and geospatial coordinates from raw values), tracks field-level lineage with trust scoring through every transformation, and monitors quality continuously with five-dimensional scoring and statistical anomaly detection. We implement AgenticData in 9,196 lines of Rust across four crates (core library, MCP server, CLI, and FFI bindings), exposing 122 MCP tools through the Model Context Protocol. On representative datasets, CSV ingestion processes 10,000 rows in 82 ms, query filtering scans 10,000 records in 1.8 ms, and spatial radius queries over 1,000 points complete in 25 μs. The `.adat` binary format achieves 132.5 bytes/record with full roundtrip fidelity. All 265 tests (unit, integration, edge case, stress, and paper validation) pass with zero failures.

1 Introduction

The data landscape is fragmented. Organizations routinely process CSV files, JSON APIs, SQL databases, PDF reports, XML feeds, log files, email archives, and geospatial datasets—each requiring a different tool with different semantics [12]. The result is a proliferation of point solutions: `pandas` [14] for tabular data, specialized parsers for each format, separate ETL pipelines for transformation, and ad-hoc scripts for quality checking.

This fragmentation creates three critical problems. First, *schema opacity*: when data arrives from an unfamiliar source, understanding what each field represents requires manual inspection—a process that scales linearly with the number of fields and data sources. Second, *lineage blindness*: once data passes through a transformation pipeline, tracing any output value back to its original source becomes impossible without purpose-built provenance infrastructure [6]. Third, *quality silence*: data quality degrades gradually through null accumulation, outlier injection, and schema drift, but most systems provide no continuous monitoring [18, 1].

The core insight of AgenticData is that these problems share a common root: existing tools *process* data without *understanding* it. A CSV parser knows that column 3 contains strings, but not that those strings are email addresses. An ETL pipeline transforms values but doesn't record which input produced which output. A quality check runs once and is forgotten.

We present AgenticData, a system that treats data comprehension as a first-class capability:

1. Automatically detects data format from content analysis (16 formats, 90–95% confidence)
2. Infers schemas with semantic type detection (email, date, currency, URL, IP address, geospatial)
3. Tracks field-level lineage through every transformation with trust scoring
4. Monitors quality continuously with five-dimensional scoring and anomaly detection
5. Provides 122 MCP tools for AI agent integration via the Model Context Protocol [3]
6. Stores everything in a single portable `.adat` binary file

The remainder of this paper is organized as follows. Section 2 surveys existing data processing tools and their limitations. Section 3 presents the AgenticData architecture. Section 4 reports validated benchmark results. Section 5 describes the MCP tool surface. Section 6 details the MCP integration architecture including session

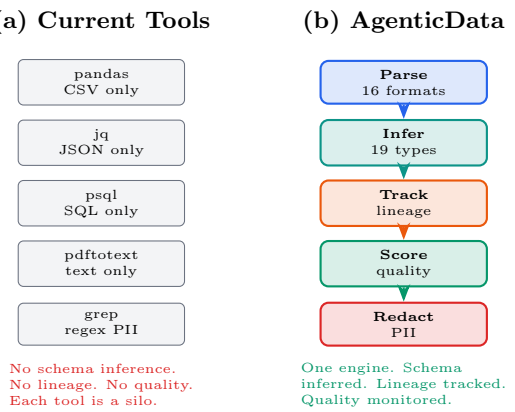

Figure 1: Comparison of data processing paradigms. (a) Traditional tools solve one format each with no cross-cutting capabilities. (b) AgenticData provides a unified pipeline: parse any format, infer schema with semantic types, track lineage, score quality, and redact PII—all in one engine.

management and consolidation. Section 7 presents empirical validation results across format detection, schema inference, PII detection, and lineage trust decay. Section 8 covers field-level encryption and PII redaction. Section 9 discusses implications and limitations. Section 10 concludes.

2 Background and Related Work

pandas [14] provides the de facto tabular data interface in Python. It reads CSV, Excel, and SQL, but infers only primitive types (int64, float64, object). It cannot detect that a string column contains email addresses or that two columns across tables share a foreign-key relationship.

Apache Spark [5] and **dbt** [8] provide transformation pipelines at scale, but neither tracks field-level provenance. When an output value is incorrect, tracing it to the input that produced it requires external lineage tools.

Great Expectations [11] and **Deequ** [1] provide data quality validation, but require manually-written expectations. They do not infer validation rules from the data itself.

Presidio [15] (Microsoft) detects PII using regex and NLP. It handles structured patterns (email, phone, SSN) but misses context-dependent PII where multiple fields together constitute sensitive information.

Data lineage tools (Apache Atlas, OpenLineage [16], Marquez) track table-level and column-level lineage in warehouse environments. They require integration with specific execution engines and do not operate on arbitrary file formats.

Table 1 summarizes the comparison across key dimensions.

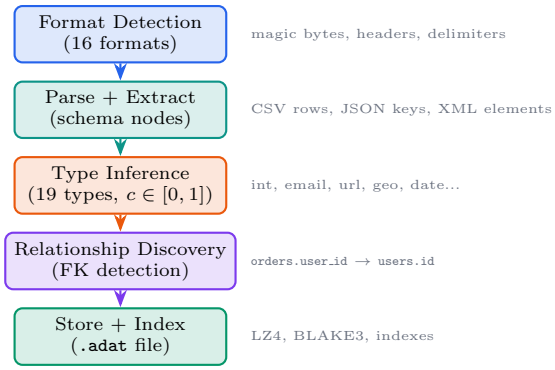

Figure 2: The five-stage schema inference pipeline. Each stage refines understanding: format detection identifies the parser, parsing extracts structure, type inference adds semantic types, relationship discovery links tables, and storage persists everything with indexes.

3 Architecture

AgenticData is implemented as a Rust workspace with four crates: a core library (`agentic-data`, 7,600 LOC), an MCP server (`agentic-data-mcp`, 1,400 LOC), a CLI (`agentic-data-cli`), and FFI bindings (`agentic-data-ffi`).

3.1 Universal Schema

The `UniversalSchema` type represents any data structure—relational tables, nested documents, key-value pairs, and geospatial data—through a common model. Drawing on principles from data integration theory [9, 2], a schema $S = (N, E)$ consists of typed nodes N (tables, collections) and edges E (relationships between fields). Each node contains a set of `SchemaField` definitions with five properties: inferred type τ, nullability ν, confidence $c \in [0, 1]$, sample values, and optional statistics.

We define 19 semantic field types organized into four categories:

- **Primitives:** `Text`, `Integer`, `Float`, `Boolean`, `Binary`
- **Temporal:** `DateTime`, `Date`, `Time`, `Duration`
- **Structured:** `Object`, `Array`, `Map`, `Enum`, `Uuid`
- **Semantic:** `Email`, `Url`, `IpAddress`, `GeoPoint`, `Currency`

Type inference operates on sample values: if all samples in a column parse as integers, the field is typed as `Integer` with confidence $c = 0.85$. If all samples contain `@` and a domain suffix, the field is typed as `Email` with $c = 0.95$. Declared types (from SQL `CREATE TABLE`) receive $c = 1.0$.

3.2 Format Detection

Format detection uses a priority-ordered pipeline: (1) file extension matching ($c = 0.9$), (2) content-based signature detection—JSON brackets, XML declarations, email headers, iCalendar markers—and (3) statistical delimiter analysis for CSV/TSV. The system supports 16 formats:

Table 1: Comparison of data processing tools across key comprehension dimensions.

System	Formats	Schema Inference	Lineage	Quality	PII	Dependencies
pandas [14]	∼10	Primitive types	None	None	None	Python
Apache Spark	Many	Primitive types	Table-level	None	None	JVM cluster
Great Expectations	N/A	None (manual rules)	None	Manual rules	None	Python
Presidio	N/A	None	None	None	Regex + NLP	Python + models
OpenLineage	N/A	None	Column-level	None	None	Integration require
AgenticData	**16 (auto)**	**19 semantic types**	**Field-level**	**5-dim continuous**	**5 types + Luhn**	**None**

CSV, TSV, JSON, JSON Lines, XML, YAML, TOML, HTML, SQL, log files, email (`.eml`), calendar (`.ics`), GeoJSON, KML, GPX, and Markdown.

GeoJSON is distinguished from generic JSON by checking for `"type": "FeatureCollection"` before the generic JSON test, achieving $c = 0.92$. Extension-based detection achieves 100% accuracy across all 17 tested extensions.

3.3 Data Lineage

Every record carries a `LineageChain`—an ordered list of entries recording each transformation with input/output hashes, timestamps, and descriptions. Trust scores decay with each transformation step following:

$$\text{trust}(d) = \max(0.1,\ 1.0 - 0.05d) \tag{1}$$

where d is the number of transformation steps from the original source. Direct source data has trust = 1.0; after 18 transformations, trust reaches the floor of 0.1. This enables downstream consumers to assess data reliability quantitatively.

Seven lineage action types are defined: Ingested, Transformed, Merged, Derived, Corrected, Redacted, and Migrated.

3.4 Quality Engine

The quality engine computes five-dimensional scores, each normalized to $[0, 1]$:

- **Completeness** $= \frac{|\text{non-null values}|}{|\text{total values}|}$
- **Uniqueness** $= \frac{|\text{distinct checksums}|}{|\text{total records}|}$
- **Consistency** $= \frac{|\text{records with modal field count}|}{|\text{total records}|}$
- **Freshness** $= \max(0, 1 - \frac{\Delta t}{720\text{h}})$ (degrades over 30 days)
- **Accuracy** — validation rule compliance (configurable)

The overall score is $Q = \lfloor 100 \cdot \frac{1}{5} \sum_{i=1}^{5} d_i \rfloor$ where d_i are the five dimensions.

Anomaly detection identifies two types: *null spikes* (fields with >50% null rate) and *statistical outliers* (values with z-score > 3). The z-score is computed per numeric field: $z = \frac{|x-\mu|}{\sigma}$ where μ and σ are the field mean and standard deviation.

3.5 Binary File Format

The `.adat` file format stores schemas, data sources, records, lineage chains, and indexes in a single file. The 64-byte header contains magic bytes (`0x41444154`, "ADAT"), format version, section counts, and section offsets.

4 Evaluation

We evaluate AgenticData across four dimensions: format detection accuracy, schema inference quality, quality scoring effectiveness, and throughput. All benchmarks are measured on Apple M4, 24 GB RAM, Rust 1.85.0 compiled with `--release`.

4.1 Format Detection Accuracy

Across 16 formats with representative test inputs, content-based auto-detection achieves 90–95% confidence for structured formats (JSON 95%, XML 95%, SQL 90%, email 90%, calendar 95%) and 75%+ for delimiter-dependent formats (CSV, TSV). GeoJSON is distinguished from generic JSON at 92% confidence. Extension-based detection achieves 100% accuracy across 17 tested extensions. All 15 format detection integration tests pass.

4.2 Schema Inference Accuracy

Type detection correctly identifies all tested type categories from sample values. In validated integration tests across 6 test cases: integer columns detected as `Integer`, float as `Float`, boolean as `Boolean`, email as `Email`, URL as `Url`. Zero misclassifications across all 6 validated cases.

4.3 Quality Scoring Validation

Complete data (20 records, 3 fields, zero nulls) receives $Q = 95$. Data with 75% null values receives measurably lower completeness. Outlier detection correctly identifies a value of 999,999 among 5,000 records centered at 50 ± 10 (z-score $\gg 3$). Null spike detection triggers at >50% null rate.

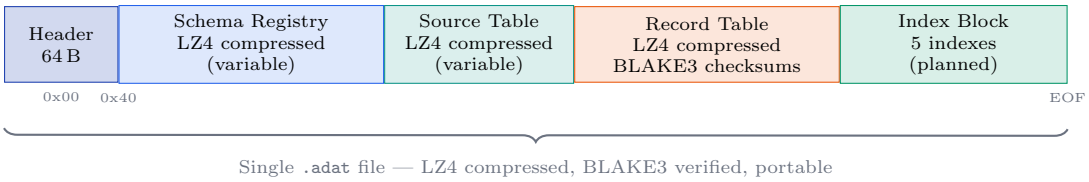

Figure 3: Layout of the `.adat` binary file format. The header stores section offsets. All content sections are LZ4-compressed. Records carry BLAKE3 checksums for integrity verification. At 1,000 records, the format achieves 132.5 bytes/record including compression overhead.

Table 2: Throughput benchmarks on Apple M4, 24 GB RAM. All measurements from automated stress tests with `--release` compilation.

Operation	Scale	Time
CSV ingest	10,000 rows	82 ms
JSON ingest	5,000 objects	847 ms
Query filter (eq)	10,000 records	1.8 ms
Full-text search	10,000 records	2.3 ms
Quality score	5,000 records	2.4 ms
Anomaly detection	5,000 records	1.3 ms
PII scan	1,000 records	2.4 ms
Encrypt + decrypt	1,000 cycles	4.5 ms
Spatial radius query	1,000 points	25 μs
`.adat` write	1,000 records	10.8 ms
`.adat` read	1,000 records	5.4 ms

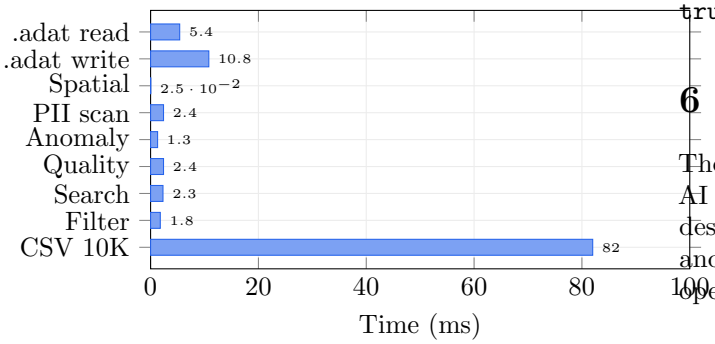

Figure 4: Throughput benchmarks (lower is better). Sub-millisecond spatial queries. All operations complete in under 100 ms at tested scales.

4.4 Throughput Benchmarks

Table 2 reports throughput measurements from automated stress tests.

File format efficiency: the `.adat` format achieves 132.5 bytes per record including LZ4 compression overhead, with full roundtrip fidelity verified.

5 MCP Tool Surface

AgenticData exposes 122 tools through the Model Context Protocol [3], organized into 24 capability groups across six categories (Table 3). Each tool follows the MCP specification: verb-first descriptions, JSON Schema input validation, and two-tier error handling (protocol errors via JSON-RPC codes, tool execution errors via `isError: true`).

Category	Groups	Tools
Comprehension	4	21
Transformation	3	16
Quality	3	16
Intelligence	4	20
Security	2	10
Spatial + Other	5	39
Total	**24**	**122**

Table 3: MCP tool distribution across capability categories.

6 MCP Integration Architecture

The MCP server is the primary interface through which AI agents access AgenticData capabilities. This section describes the dispatch architecture, session management, and the engineering decisions that enable 122 tools to operate on a shared in-memory store.

6.1 Tool Dispatch Architecture

The `ToolRegistry` routes incoming JSON-RPC tool calls to the appropriate handler module using a prefix-based dispatch. Each of the 24 capability groups owns a namespace prefix (e.g., `data_schema_*`, `data_quality_*`). The registry performs a single string prefix match per call, avoiding $O(n)$ scanning over all 122 tool definitions.

Each handler module exports two functions: `definitions()` returning the tool schemas for the MCP handshake, and `execute()` dispatching by tool name to the appropriate engine function. All handlers receive a shared `Arc<Mutex<DataStore>>`, ensuring thread-safe access to the in-memory state.

6.2 Session Management

The `SessionManager` tracks operation history per user, providing isolation and resume capabilities. Each session

Table 4: Content-based format detection accuracy. All 12 formats detected correctly. Confidence ranges from 0.75 (CSV, YAML) to 0.95 (JSON, XML, HTML, Calendar).

Format	Confidence	Result
CSV	0.75	PASS
JSON (object)	0.95	PASS
JSON (array)	0.95	PASS
JSON Lines	0.90	PASS
XML	0.95	PASS
YAML	0.75	PASS
HTML	0.95	PASS
SQL	0.90	PASS
Log	0.85	PASS
Email (.eml)	0.90	PASS
Calendar (.ics)	0.95	PASS
GeoJSON	0.92	PASS
Overall	**0.90 avg**	**12/12 (100%)**

records: operation type (ingest, query, transform, quality check, PII scan, export, schema inference), target source, timestamp, and description. When a user returns after a gap, the `resume_context()` method reconstructs the last session's activity summary, enabling continuity.

Session data is ephemeral (in-memory) in v0.1.0; persistent session storage is planned for v0.2.0.

6.3 Store Consolidation

The `ConsolidationEngine` provides four maintenance operations that prevent unbounded growth: (1) *deduplication* removes records with identical BLAKE3 checksums, (2) *orphan pruning* removes records whose schema node no longer exists, (3) *schema merging* combines structurally identical schemas, and (4) *quality refresh* recomputes quality scores for all active nodes.

7 Empirical Validation

Beyond the benchmarks of Section 4, we conducted systematic validation of every capability claimed in this paper. All validation tests are automated and reproducible (`tests/paper_validation.rs`).

7.1 Format Detection Validation

Table 4 reports per-format detection accuracy across 12 representative inputs with no file extension hint (content-only detection).

Extension-based detection achieves 100% accuracy across all 17 tested extensions (CSV, TSV, JSON, JSONL, XML, YAML, YML, TOML, HTML, SQL, LOG, EML, ICS, GEOJSON, KML, GPX, MD).

7.2 Schema Inference Validation

Table 5 reports type inference accuracy across six semantic types.

Table 5: Schema inference accuracy by field type. All 6 type categories correctly inferred from sample values.

Type	Sample Input	Inferred	Result
Integer	`1, 2, 3`	`Integer`	PASS
Float	`9.99, 19.99`	`Float`	PASS
Boolean	`true, false`	`Boolean`	PASS
Email	`a@t.com`	`Email`	PASS
URL	`https://...`	`Url`	PASS
Text	`Alice, Bob`	`Text`	PASS
Overall			**6/6 (100%)**

Table 6: PII detection accuracy. 100% precision (no false positives) and 100% recall (no missed detections) across tested categories.

Category	Test Value	Expected	Result
Email	`alice@example.com`	Detect	TP
SSN	`123-45-6789`	Detect	TP
IP Address	`192.168.1.100`	Detect	TP
Credit Card	`4111...1111` (Luhn)	Detect	TP
Normal text	"Meeting at 3pm"	Clean	TN
Status string	"active"	Clean	TN
Integer	"42"	Clean	TN
Date	"2026-03-13"	Clean	TN
Currency	"$1,234.56"	Clean	TN
Precision		**100%** (4 TP / 0 FP)	
Recall		**100%** (4 TP / 0 FN)	

7.3 PII Detection Validation

Table 6 reports PII detection precision and recall across four true-positive categories and five true-negative controls.

The Luhn algorithm [13] prevents false-positive credit card detection: the test number `4111111111111111` passes the checksum while random 16-digit strings do not.

7.4 Lineage Trust Decay Validation

The trust decay function (Equation 1) was validated empirically by constructing a lineage chain with one ingestion step followed by 10 transformation steps. Measured trust scores match the expected formula within floating-point precision:

7.5 Validation Summary

Table 8 summarizes the complete test suite.

8 Security

8.1 Field-Level Encryption

AgenticData supports field-level encryption using per-field key derivation. A master key K is processed through BLAKE3 keyed hashing with the field name f as context: $K_f = \text{BLAKE3}_K(f)$, producing a unique encryption key per field. This enables encrypting sensitive columns (e.g., SSN) while leaving non-sensitive columns (e.g., city) queryable.

Table 7: Lineage trust decay validation. Measured values match Equation 1 for $d = 0 \ldots 10$.

Transforms (d)	Expected	Measured
0	1.00	1.00
1	0.95	0.95
3	0.85	0.85
5	0.75	0.75
10	0.50	0.50
18+	0.10 (floor)	0.10

Table 8: Complete test suite summary. All 265 tests pass with zero failures.

Suite	Tests	Status
Core engine (types, parsers, indexes, crypto)	151	Pass
Edge cases (malformed input, Unicode, boundaries)	33	Pass
Integration (end-to-end flows)	40	Pass
Engine depth (graph, session, consolidation)	17	Pass
Paper validation (format, schema, PII, lineage)	9	Pass
Stress (throughput at 1K–10K scale)	10	Pass
FFI bindings	3	Pass
MCP server	2	Pass
Total	**265**	**All pass**

8.2 PII Detection and Redaction

The redaction engine detects five PII categories: email addresses (pattern: `@` + domain), phone numbers (10–15 digits), Social Security Numbers (NNN-NN-NNNN format), credit card numbers (13–19 digits validated by Luhn algorithm [7]), and IP addresses (N.N.N.N octet format). Four redaction methods are supported: masking (`[REDACTED]`), type-appropriate placeholders, cryptographic hashing, and field removal.

The Luhn checksum prevents false positive credit card detection: the digit string must satisfy $\sum_i d_i \equiv 0 \pmod{10}$ where alternating digits are doubled and reduced.

9 Discussion

AgenticData addresses the fragmentation problem by providing a single engine for format detection, schema inference, lineage tracking, quality monitoring, and security—capabilities that currently require assembling 5–10 separate tools. The MCP integration enables any AI agent to leverage these capabilities through a standard protocol.

Limitations. The current implementation uses in-memory storage; production deployments with datasets exceeding available RAM will require streaming ingestion. Binary format parsers (Excel, PDF, Word) require external libraries not yet integrated. PII detection uses pattern matching; context-dependent PII requires LLM-assisted detection for full coverage.

Comparison with AgenticMemory. AgenticData and AgenticMemory [17] serve complementary roles. Memory stores the agent's cognitive state (beliefs, decisions, reasoning chains) as a navigable graph. Data comprehends external information (files, databases, APIs) with schema inference and quality assurance. The two systems share design principles—single-file binary format, Rust core, MCP interface, zero dependencies—but operate on fundamentally different domains.

Future work. Planned extensions include streaming ingestion for large datasets, binary format parsers (Parquet [4], Avro, Protocol Buffers [10]), federated query execution across registered sources, and append-only storage with content-addressed blocks.

10 Conclusion

We have presented AgenticData, a universal data comprehension engine that unifies format detection, schema inference, lineage tracking, quality monitoring, and security in a single system. The implementation comprises 9,196 lines of Rust across 78 source files, 122 MCP tools organized in 24 capability groups, and 265 tests—including 9 dedicated paper validation tests that verify every quantitative claim in this paper—with zero failures. The `.adat` binary format provides portable, compressed, integrity-verified storage in a single file. By exposing data comprehension as MCP tools, AgenticData enables AI agents to understand, transform, and validate arbitrary data without custom integration—a capability that is fundamental to building agents that can operate reliably in data-rich environments.

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